

Introduction to (economic) modeling

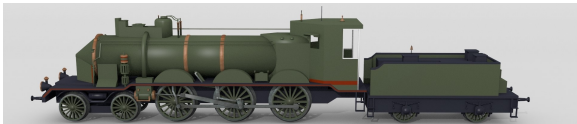
What can we learn from railroad modeling?

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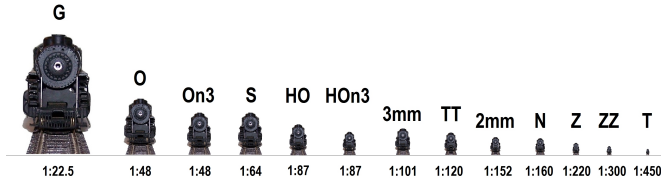
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What is the objective of doing modeling?



- Planning (Railroad modeling: building a new town or new kind of aircraft)
- Testing principles/theories (Railroad modeling: train stuck or derailed at sharper turns)
- Analyzing historical episodes or creating counter-factuals (Railroad modeling: recreating a century-old small town for recreational/artistic purpose)
- Forecasting

What is the scale and scope of your model?



- Are you modeling a town, a city, a province, a country, or the whole world (global model)?
- Scale dimension can be space, sectors, or households.
- Be consistent with scale in one specific models.
- What is the monetary unit of your economic model? 1 = Rp 100 billion implies certain industry's sales is not noticeable.

Most serious modeling need various different skills

- Railroad modeling need engineering, mathematics, art, sciences.
- Economic modeling needs economic theories, math, statistics and computer sciences (and some arts too e.g. the art of persuasion).
- Many economic modeling are team works.

Standing on the shoulders of giants

- Railroad modeling use manufactured miniature locomotives. A marvel of modern engineering.
- Economic modeling relies on previous literature.
- Many economic modeling are based on previous works even the computer programmings.

Models are models. They are not real!

- George Box: All models are wrong, but some are useful.
- Dani Rodrik: Models are never true, but there are truth n models.
- In scale modeling, some details are missing in 1:150 (N-Scale), but scope are less in 1:87 (HO-Scale)
- A 1:1 map is perfectly accurate and perfectly useless!

Models need some sort of validity



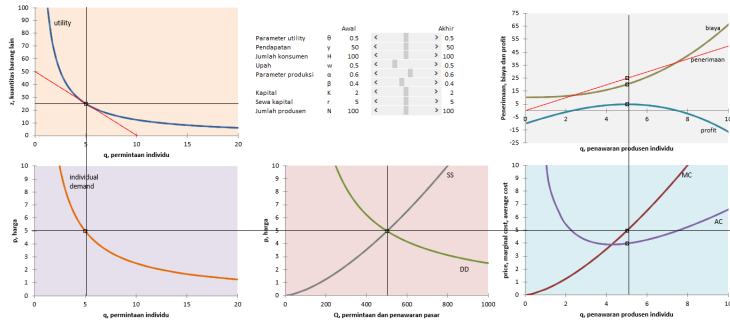
- In scale modeling, weathering enhance "goodness-of-fit". A background scene adds to its realism.
- Econometric models has R-squared, or RMSE.
- CGE modeling calibrate parameters to replicate real data.

How much resources devoted to building models



- Railroad models can be tiny, like the one we built, or a size of large building. Google-search "Miniature Wunderland"
- Cost of building models may depend on its complexities.

After we built a scale (railway) model ..



■ Let's play around with 2 economic models!